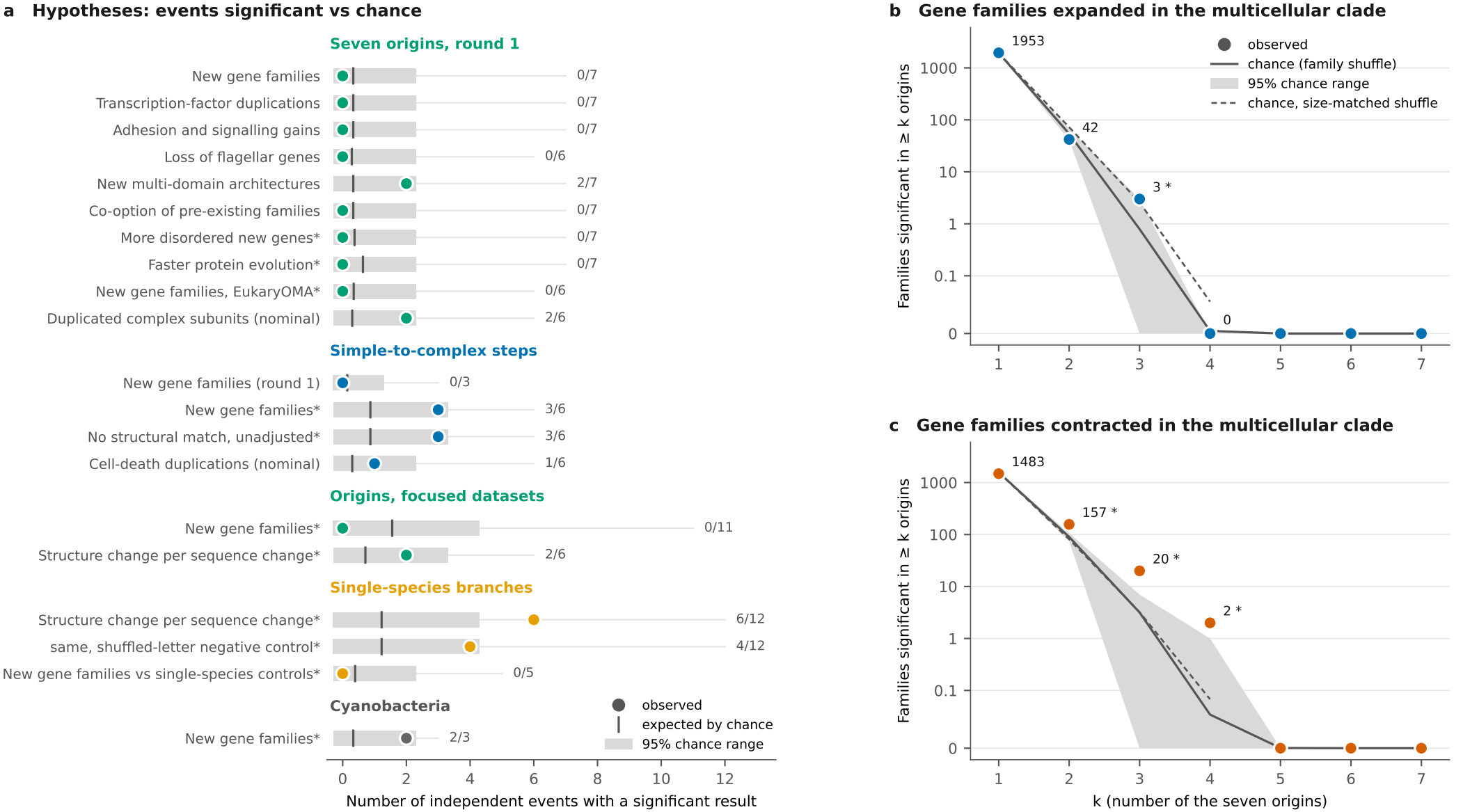
a Hypotheses: events significant vs chance
Seven origins, round 1
New gene families
Transcription-factor duplications
Adhesion and signalling gains
Loss of flagellar genes
New multi-domain architectures
Co-option of pre-existing families
More disordered new genes*
Faster protein evolution*
New gene families, EukaryOMA*
Duplicated complex subunits (nominal)
0/7
0/7
0/7
0/6
2/7
0/7
0/7
0/7
0/6
2/6
Simple-to-complex steps
New gene families (round 1)
New gene families*
No structural match, unadjusted*
Cell-death duplications (nominal)
0/3
3/6
3/6
1/6
Origins, focused datasets
New gene families*
Structure change per sequence change*
0/11
2/6
Single-species branches
Structure change per sequence change*
same, shuffled-letter negative control*
New gene families vs single-species controls*
6/12
4/12
0/5
Cyanobacteria
New gene families*
2/3
observed
expected by chance
95% chance range
0
2
4
6
8
10
12
Number of independent events with a significant result
b Gene families expanded in the multicellular clade
Families significant in ≥ k origins
1000
100
10
1
0.1
0
1953
42
3 *
0
observed
chance (family shuffle)
95% chance range
chance, size-matched shuffle
1
2
3
4
5
6
7
c Gene families contracted in the multicellular clade
Families significant in ≥ k origins
1000
100
10
1
0.1
0
1483
157 *
20 *
2 *
1
2
3
4
5
6
7
k (number of the seven origins)